B.Tech (Computer Engineering) 5th Semester Examination Session 2025-26
Software Engineering (CSC- 504)
End Semester Examination

Time: 03 Hours **Max Marks:45**

Note:

1. M-Marks, BL- Blooms Taxonomy Level, CO- Course Outcome
2. Electronic gadgets are not allowed in the examination hall like mobile phone, smart watch etc.
3.Students will only be allowed to use non-programmable scientific calculator.
4. Attempt any two parts from each question. All questions carry equal marks

S.No		Question Statement	M	BL	CO
Q1	a.	Write down the difference between Program & software with suitable example	4.5		1
	b.	Explain the Spiral Model of software development. What are the major phases in each spiral cycle.	4.5	L2	1
	c.	What is the Capability Maturity Model (CMM) of Software Development? Explain all five levels of the CMM with their key characteristics.	4.5	L2	1
Q2.	a.	A software project is estimated at 55 KLOC (thousand lines of code). Using the **COCOMO model** in the **Embedded mode**, and assuming the Effort Adjustment Factor (EAF) = 1.25, calculate the following: 1. Effort 2. Development Time 3. Team Size 4. Productivity	4.5	L3	2
	b.	List and explain the features of a good Software Requirements Specification (SRS).	4.5	L2	2
	c.	What is Risk Management in Software Engineering? Write down the steps of Risk Management.	4.5	L2+L3	2
Q3	a.	Draw and explain the Data Flow Diagram (DFD) for a Library Management System, showing both Level 0 and Level 1 diagrams.	4.5	L3	3
	b.	Define **cohesion** and explain the **different types of cohesion** with suitable examples.	4.5	L2	3
	c.	Develop an **Activity Diagram** for an **Online Examination System**, showing the flow from student login to result generation.	4.5	L3	3
Q4	a.	Explain different **levels of software testing** — Unit, Integration, System, and Validation Testing — with suitable examples.	4.5	L2	4
	b.	**Scenario:** A system accepts age input between 18 and 60. • Input Range: 18 to 60 • Valid inputs: 18–60 • Invalid inputs: less than 18 or greater than 60 **Explain the following functional testing techniques with above scenario–based test cases:** **a) Boundary Value Analysis** **b) Equivalence Class Partitioning**	4.5	L2+L3	4
	c.	Explain black box and white box testing with suitable example	4.5	L2	4
Q5	a.	Describe various **types of software maintenance** with suitable examples for each.	4.5	L2	5
	b.	Explain the **process of reverse engineering** in software engineering. Illustrate the steps involved with a neat labelled diagram.	4.5	L2	5
	c.	Explain the **working process of the Quick-Fix Model** with diagram. List any two **advantages** and **disadvantages** of this model.	4.5	L3	5